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| **Fiscal Year:** | FY 2004 | **Task Last Updated:** | FY 12/02/2004 |
| **PI Name:** | Almeida-Porada, Graca M.D., Ph.D. | | |
| **Project Title:** | Stem Cell Plasticity Under Simulated Microgravity | | |
| **Division Name:** | Physical Sciences | | |
| **Program/Discipline:** | BIOTECHNOLOGY | | |
| **Program/Discipline--Element/Subdiscipline:** | BIOTECHNOLOGY--Cellular biotechnology & tissue engineering | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | None | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
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| **Zip Code:** | 89557-0104 | **Congressional District:** | 2 |
| **Comments:** | | | |
| **Project Type:** | Ground | **Solicitation / Funding Source:** | 00-HEDS-03 |
| **Start Date:** | 08/01/2001 | **End Date:** | 09/30/2005 |
| **No. of Post Docs:** | 0 | **No. of PhD Degrees:** | |
| **No. of PhD Candidates:** | 2 | **No. of Master' Degrees:** | |
| **No. of Master's Candidates:** | 1 | **No. of Bachelor's Degrees:** | |
| **No. of Bachelor's Candidates:** | 1 | **Monitoring Center:** | NASA JSC |
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| **Flight Program:** | | | |
| **Flight Assignment:** | | | |
| **Key Personnel Changes/Previous PI:** | | | |
| **COI Name (Institution):** | | | |
| **Grant/Contract No.:** | NAG9-1340 | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |
| | The intricate process of embryonic development initiates with a single totipotent stem cell that by undergoing multiple cycles of division/differentiation gives rise to an entire multicellular organism. Until recently it was thought that with each subsequent division, the progeny of these original stem cells shed some of their totipotency and began committing to a certain tissue and/or cellular lineage. Recent studies have shown that multipotent stem cells persist throughout life and participate in normal replacement and repair within their respective tissues. Furthermore, results from both in vitro and in vivo studies have now suggested that stromal/mesenchymal stem cells from bone marrow (BM-MSC) may harbor the ability to de-differentiate and reprogram themselves to give rise to cells of unrelated lineages. We have recently begun to use our laboratory's unique human-sheep xenograft model system of in utero human stem cell transplantation to explore the plasticity that appears to exist within the primitive stem cell population present within several organs. We have now shown that human neural stem cells are able to produce both differentiated blood cells and hematopoietic stem/progenitor cells with the ability to engraft the hematopoietic systems of secondary transplant | | |

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| **Task Description:** | recipients. We have also demonstrated the formation of human hepatocytes following hematopoietic stem cell, mesenchymal stem cell, or neural stem cell transplantation in utero. Since blood, brain, and liver have not yet been shown to be capable of interconverting in vitro, we concluded that the influence of the fetal microenvironment was responsible for unlocking the hidden potential of these organ-specific stem cells, enabling them to trans-differentiate and give rise to cells of other tissues. Indeed, the fetus represents a unique system in which to assay stem cell differentiation potential and plasticity because of both the wide array of inductive environmental signals and the need for multiple cell types created by the exponential growth rate of a developing fetus. However, based on the rapidly accumulating data that conditions of microgravity alter the gene expression of cultured cells, we wish in the present proposal to explore an alternate explanation for the plasticity of the stem cell(s) within our model system. In addition to the afore-mentioned characteristics, the fetus also represents a unique biological system because it is suspended in a fluid-filled sac and thus approaches weightlessness, at least at the early stages of development. Thus, we propose that the weightless state of the early gestational fetus is also a contributing factor to the observed plasticity within the stem cell compartment of various organ systems. While a fetal transplantation model system is ideal for evaluating many aspects of stem cell plasticity, the ability to address these issues in vitro would allow dissection and manipulation of the processes controlling the interconversion of cells from seemingly disparate embryologic origins. Unfortunately, to date, it has not been possible to perform studies such as these in vitro due to the difficulties with accurately recreating each organ's microenvironment in traditional monolayer culture systems. The development of NASA's rotating wall vessel (RWV) has now allowed the successful long-term culture of both mixed cell populations that have been largely refractory to standard culture approaches and relatively large tissue fragments from a variety of organs, maintaining normal cytoarchitecture. Thus, the overall goal of these studies is to use the RWV to begin addressing the complex question of mesenchymal stem cell plasticity in vitro while simultaneously evaluating the impact that gravity (or its relative absence) has on the inherent plasticity of stem cells and their expression of the gene pathways responsible for this malleability. The use of the RWV will allow the performance of carefully designed in vitro studies employing clonally derived retroviral marked populations of BM-MSC that will enable us to answer the difficult question of whether these stem cells constitute a mixed population, each member of which is committed to its own embryonic fate, or whether these stem cells are truly pluripotent. We will perform studies in which 1) BM-MSC are cultured alone under conditions designed to induce differentiation into cells of liver, muscle, bone, cartilage, endothelial, epithelial, and neural derivation, 2) BM-MSC are cultured in each of these conditions in the presence of cellular suspensions prepared from each of the afore-mentioned tissues, and 3) BM-MSC are cultured in each of these conditions in the presence of tissue fragments from each of the afore-mentioned organs. By including studies with BM-MSC alone, with mixed cell types, and with tissue fragments, we will be able to examine BM-MSC plasticity in vitro at the single cell, multi-cell (spheroid), and tissue (organoid) levels. The studies at the single cell level will also address the important issue of the impact of microgravity on cellular proliferation and telomere/telomerase homeostasis. Once these in vitro studies have been completed, we will then use our unique human-to-sheep model of in utero transplantation to evaluate whether culture under conditions of microgravity impacts on the in vivo engraftment ability and plasticity/differentiative potential of BM-MSC. The identification of a population of cells that is easily obtained and readily available and that possesses true stem cell plasticity, could potentially be of clinical utility in the treatment of multiple diseases in various organs. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | |
| **Task Progress:** | We have previously reported that Stro-1 is able to identify populations of human MSCs in different organs. Here, we investigated whether microgravity can alter the “stemness” of MSCs in such a way as to hinder the ability of these cell populations to differentiate into various cells/lineages. As the effects of microgravity on bone loss are well known, we investigated whether alterations in MSC’s differentiative pathways could lead to osteoporosis. Thus, we compared the ability of MSC to differentiate into adipocytes and bone under regular or microgravity culture conditions. We used Ca2+ content, Von Kossa staining and tetracycline incorporation to determine osteocytic and Oil-Red-O staining for adipocytic differentiation, respectively. Adipocytes and osteocytes were obtained within 9 and 12 days, in regular cultures. Under microgravity conditions, adipocytic differentiation occurred in 12 days, similar to that of regular cultures. However, osteocytic differentiation under microgravity conditions took 4 times longer than cultures grown under normal gravity . Since MSC give rise to both osteoblasts and adipocytes, it is possible that an imbalance in these differentiative pathways may lead to impaired osteogenesis, contributing to osteoporosis. We also investigated whether microgravity affected MSC differentiation into other cell lines. To this end, we grew MSCs for 5 days under microgravity conditions and transplanted them into fetal sheep at a concentration of 1x106 cells/fetus and compared the in vivo differentiative potential of these cells with control sheep transplanted with MSC grown in regular cultures. Flow cytometric evaluation of the recipients at 60 days post-transplant for the presence of human-derived blood cells revealed that although MSCs grown under microgravity conditions generated multilineage blood cells, they did so at a lower level than their regular culture |
| **Bibliography Type:** | Description: (Last Updated: 08/28/2021) |
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